

Development of Metastatic Disease After Enrollment in the COMS Trials for Treatment of Choroidal Melanoma

Collaborative Ocular Melanoma Study Group Report No. 26

*Collaborative Ocular Melanoma Study Group**

Objective: To describe the time between treatment for choroidal melanoma and first diagnosis of metastatic disease, sites of metastasis, treatments for metastasis, and time between diagnosis of metastasis and death.

Design: Prospective, longitudinal follow-up of patients diagnosed with choroidal melanoma who were enrolled in 2 randomized trials conducted by the Collaborative Ocular Melanoma Study Group.

Methods: Systemic and laboratory evaluations were performed during follow-up according to a standard protocol for 2320 patients enrolled in the Collaborative Ocular Melanoma Study trials without evidence of melanoma metastasis or other primary cancer at baseline.

Results: Seven hundred thirty-nine patients were diagnosed with at least 1 site of metastasis during follow-up after treatment for choroidal melanoma. Five- and 10-

year cumulative metastasis rates were 25% (95% confidence interval, 23%-27%) and 34% (95% confidence interval, 32%-37%), respectively. Liver was the most common site (89%). The death rate following the report of melanoma metastasis was 80% at 1 year (95% confidence interval, 77%-83%) and 92% at 2 years (95% confidence interval, 89%-94%). Overall survival after metastasis did not vary by baseline size of primary tumor nor treatment for metastasis (when known). Long-term survival after diagnosis of metastasis was uncommon; only 8 patients survived 5 or more years.

Conclusion: Metastasis rate increased significantly with increasing primary tumor dimensions at time of patient enrollment. Prognosis after metastatic disease remains poor. Effective methods are needed to prevent, diagnose, and treat metastasis from choroidal melanoma.

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PREVIOUS STUDIES HAVE DOCUMENTED that disease-related mortality is high for patients with choroidal melanoma,¹⁻³ with liver as the most common site of metastatic disease. The time between diagnosis of metastasis and death is typically short (6-12 months), and treatment is generally ineffective for patients with metastatic disease.⁴ Long-term survival in patients diagnosed with choroidal melanoma metastasis appears to be rare and has not been systematically studied. A recent study by Kujala et al⁵ suggested that metastatic melanoma was the leading cause of death throughout a follow-up period extending for 35 years. Metastasis beyond 25 years from time of treatment is unusual and has been described in only a few case reports. Retrospective reviews of patient medical records and death registries have been fraught with patient and information selection biases. Few

studies have had sufficient data on time to metastasis, associated sites, and subsequent time to death after diagnosis of metastasis.

The Collaborative Ocular Melanoma Study (COMS) was designed as 2 randomized trials to evaluate forms of radiotherapy for choroidal melanoma with respect to overall survival and metastasis-free survival.^{2,3,6} The COMS provides the largest series of patients to date with large- or medium-sized choroidal melanoma who had no other clinically detectable cancer present at the time of treatment and who were evaluated prospectively, enrolled, treated, and followed up in a standard fashion. We have reported the COMS methods for coding metastatic and disease status of patients at the times of their deaths.⁷ Recently, an unusual COMS case was reported about a patient who had survived more than 5 years after diagnosis of liver metastasis.⁸

This report provides an update of our previous description of metastatic and disease status among patients in the COMS trial of preenucleation radiation⁹ for large choroidal melanoma and adds data for patients in the COMS trial of iodine-125 brachytherapy for medium-sized melanoma. Our primary aim was to describe the sites of metastasis, time to diagnosis of metastasis, and subsequent survival of patients within the COMS over 5 to 10 years of follow-up in these 2 randomized trials. A secondary aim was to report types of treatments administered subsequent to diagnosis of metastasis.

METHODS

The COMS design and methods have been described.^{6,7} Patients were screened, enrolled, treated as assigned randomly, and followed up at 43 clinical centers in the United States and Canada. Study characteristics of patients^{10,11} and mortality findings by treatment arm^{2,3,12} have been published. Institutional review boards of all participating institutions reviewed and approved the COMS protocol and patient consent forms. All patients gave written consent before enrollment and randomization.

Prior to enrollment, each patient was evaluated for eligibility, including specific procedures to rule out metastatic disease or another malignancy. This baseline cancer-oriented systemic evaluation was performed by a medical oncologist, radiation oncologist, or other specialist. The evaluation included a general physical examination, chest x-ray, liver function tests, and follow-up with diagnostic tests whenever there was evidence of cancer outside the eye.

After enrollment, each patient was required to have an annual medical evaluation similar to that performed at the baseline visit. Additional medical visits were scheduled to evaluate symptoms as required for good patient care. Whenever a new metastatic melanoma or primary cancer was diagnosed on the basis of any diagnostic study, a report was submitted by the study coordinator at the clinical center to the COMS Coordinating Center (Baltimore, Md) along with copies of reports of all procedures that were performed to support the diagnosis. Biopsy or cytology slides or blocks of embedded tissue were sent to the COMS Pathology Center (Madison, Wis) for histopathologic review. Diagnosis of metastasis as reported by the clinical centers was confirmed by the COMS Mortality Coding Committee (MCC) when the results of the histopathologic review were available for deceased patients. A new metastasis also could be identified for the first time during the review by the MCC.^{7,9} For this report, the tests used to confirm metastasis were obtained from the clinical center reports submitted at the time of diagnosis of metastasis. In the infrequent case that a report of metastasis had not been submitted previously from the clinical center, this information was obtained from materials reviewed by the MCC after the death of a patient. The sites of metastasis were obtained from the MCC review for decedents; otherwise, the sites were obtained from the report of metastasis submitted by the clinical center. For the subgroup of patients with metastasis diagnosed after July 10, 1995, the dates and types of treatments for metastasis were reported.

Differences in distributions of characteristics between groups of patients were compared using the χ^2 statistic, *t* test, or Wilcoxon rank sum test, as appropriate. Kaplan-Meier estimates of time from enrollment to diagnosis of metastasis as well as time from diagnosis of metastasis to death were calculated along with 95% confidence intervals (CIs).¹³ The log-rank test¹⁴ was used to compare time to event between tumor size subgroups defined for analysis

purposes as (1) longest basal diameter (LBD) less than 16 mm and apical height (HT) less than or equal to 8 mm (*n*=1199), (2) LBD less than 16 mm and HT greater than 8 mm (*n*=293), and (3) LBD greater than or equal to 16 mm (*n*=828). We used SAS statistical software (SAS Institute Inc, Cary, NC) for all statistical analysis. This report is based on data for COMS patients received at the COMS Coordinating Center by the end of the scheduled clinical follow-up for study purposes on July 31, 2000, and July 31, 2003, for the trials of preenucleation radiation and iodine-125 brachytherapy, respectively, as documented in the COMS database as of February 9, 2004.

RESULTS

RATE OF METASTASIS

Of 2320 patients enrolled in COMS randomized trials, 739 patients were diagnosed with at least 1 site of melanoma metastasis during follow-up after primary treatment for choroidal melanoma. Overall Kaplan-Meier estimates of 2-, 5-, and 10-year metastasis rates were 10% (95% CI, 9%-12%), 25% (95% CI, 23%-27%), and 34% (95% CI, 32%-37%), respectively. The rate of diagnosis of metastasis differed among tumor dimensions subgroups based on LBD and HT (**Figure 1**). The proportion of patients with metastasis increased significantly with increased baseline tumor dimensions; the difference persisted throughout follow-up. The estimated proportion with metastasis at 5 years by subgroup was 13% (95% CI, 11%-15%), 23% (95% CI, 18%-28%), and 43% (95% CI, 40%-46%) for tumors with (1) LBD less than 16 mm and HT less than or equal to 8 mm, (2) LBD less than 16 mm and HT greater than 8 mm, and (3) LBD greater than or equal to 16 mm, respectively. Similarly, the proportion with metastasis at 10 years was 21% (95% CI, 18%-23%), 35% (95% CI, 29%-41%), and 55% (95% CI, 51%-59%) for the 3 respective subgroups. There were no statistically significant differences in the distribution of baseline age by tumor size subgroup.

The median time from diagnosis of metastasis to death was less than 6 months (**Figure 2**). The death rate was 80% (95% CI, 77%-83%) and 92% (95% CI, 89%-94%) at 1 and 2 years following the report of melanoma metastasis. There was no statistically significant difference among the tumor dimensions subgroups with respect to time from diagnosis of metastasis to death (*P* = .46, log-rank test).

SITES OF METASTASIS

Tests confirming recurrence (metastasis) included computed tomography, magnetic resonance imaging, and other imaging scans (76%); biopsy (64%); liver function tests (40%); and physical or clinical examination (39%). The distributions of sites of metastasis are shown in the **Table** and were similar across baseline tumor size subgroups. The liver was the predominant site of metastasis, reported in 660 (89%) patients.⁹ Other common sites included lung (29%), bone (17%), skin and subcutaneous tissue (12%), and lymph node (11%). The number of sites documented per patient was 1 (53% of patients),

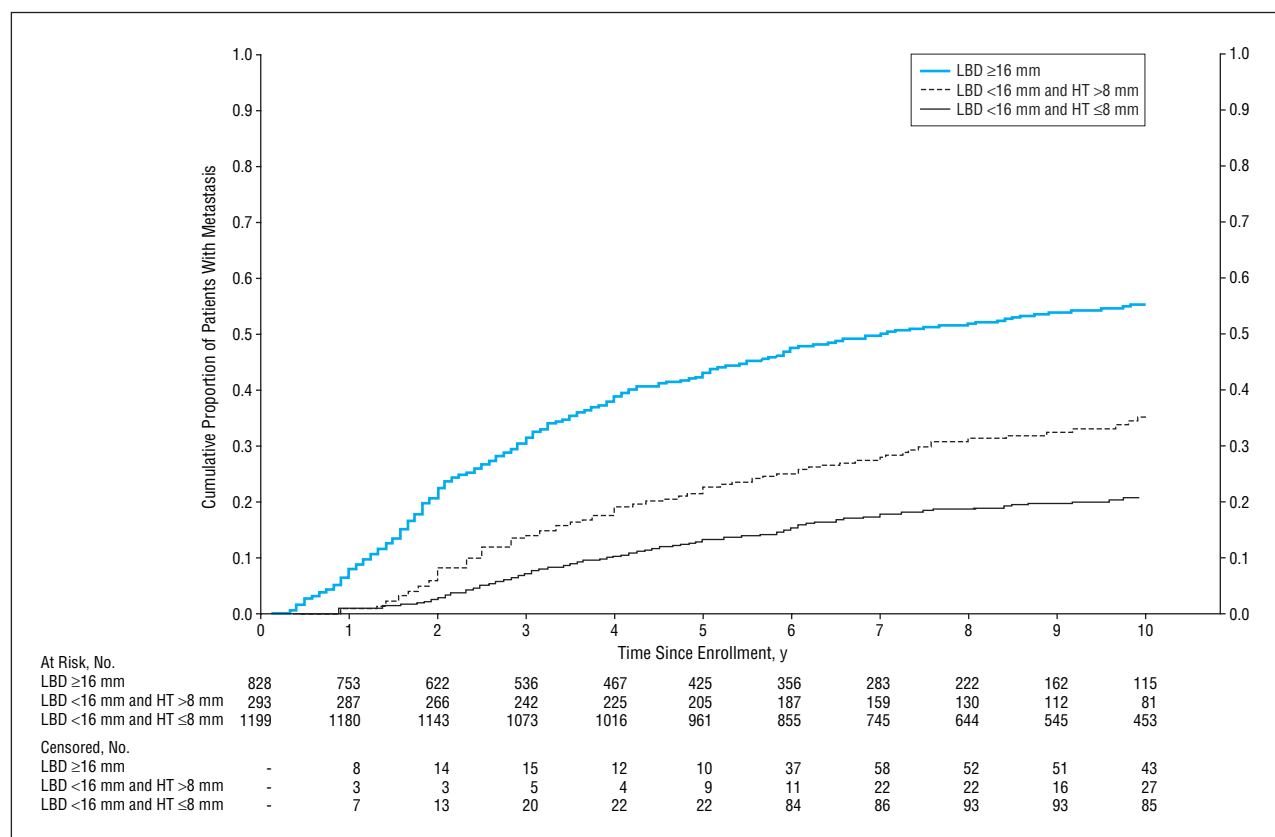


Figure 1. Time to diagnosis of first metastasis by baseline tumor size subgroup. HT indicates apical height of tumor; LBD, longest basal diameter of tumor.

2 (24% of patients), 3 (12% of patients), and 4 to 7 sites (10% of patients). For 344 (46%) patients, liver was the only site in which metastases was detected, 316 (43%) patients had metastases diagnosed in the liver plus other sites, and 79 (11%) had metastases diagnosed in sites other than the liver. This distribution was similar across tumor size subgroups.

Of the 739 patients with metastasis reported, 702 had died by the end of COMS clinical follow-up; metastasis was not discovered until the time of death in only 13 patients. All 702 deaths among patients with an earlier diagnosis of metastasis were reviewed by the COMS MCC. Among these, 561 (80%) patients were coded as dying with histopathologically confirmed melanoma metastasis. An additional 140 (20%) were coded as death with suspected melanoma metastasis based on available clinical tests and procedures, although histopathologic information was not available. There was insufficient available clinical information for MCC classification in only 1 case.

TREATMENT FOR METASTASIS

Information on planned treatment for metastasis was added to the COMS data collection form for the clinical report of metastasis on July 10, 1995. However, this information was available for a majority of cases of metastasis: 650 (88%) of the 739 total patients reported to have had metastatic melanoma prior to death. At the time of the report of metastasis, melanoma-specific treatment was not undertaken for the majority of this subset, 397 (61%)

patients. Of the remainder, 146 (23%) received chemotherapy alone, 28 (4%) radiation alone, 11 (2%) immunotherapy alone, and 38 (6%) other treatments, and 30 (5%) had combinations of chemotherapy, radiation, immunotherapy, and other therapeutic regimens. The distribution of management options did not differ significantly by baseline tumor size subgroup. The frequency and types of therapies varied by clinical center; the percentage of metastatic patients per center who received treatment ranged from 0 to 100%.

Patients treated for metastasis were significantly younger at both baseline and at the time of diagnosis of metastasis; the median age at metastasis was approximately 64 years vs 68 years for the treated and untreated groups, respectively. The baseline tumor dimensions in the treated vs untreated groups were not statistically significantly different. After adjusting for difference in age at metastasis, there were no statistically significant differences in survival after diagnosis of metastasis between the treated and untreated groups.

LONG-TERM SURVIVORS WITH METASTASIS

Patients who survived longer than 6 months after metastasis was diagnosed were significantly younger (median age of 60 years) than patients whose time between diagnosis of metastasis and death was shorter than 6 months (median age of 65 years). They otherwise did not differ by baseline tumor size, sex, or family history of cancer. A total of 8 patients survived 5 years or more after

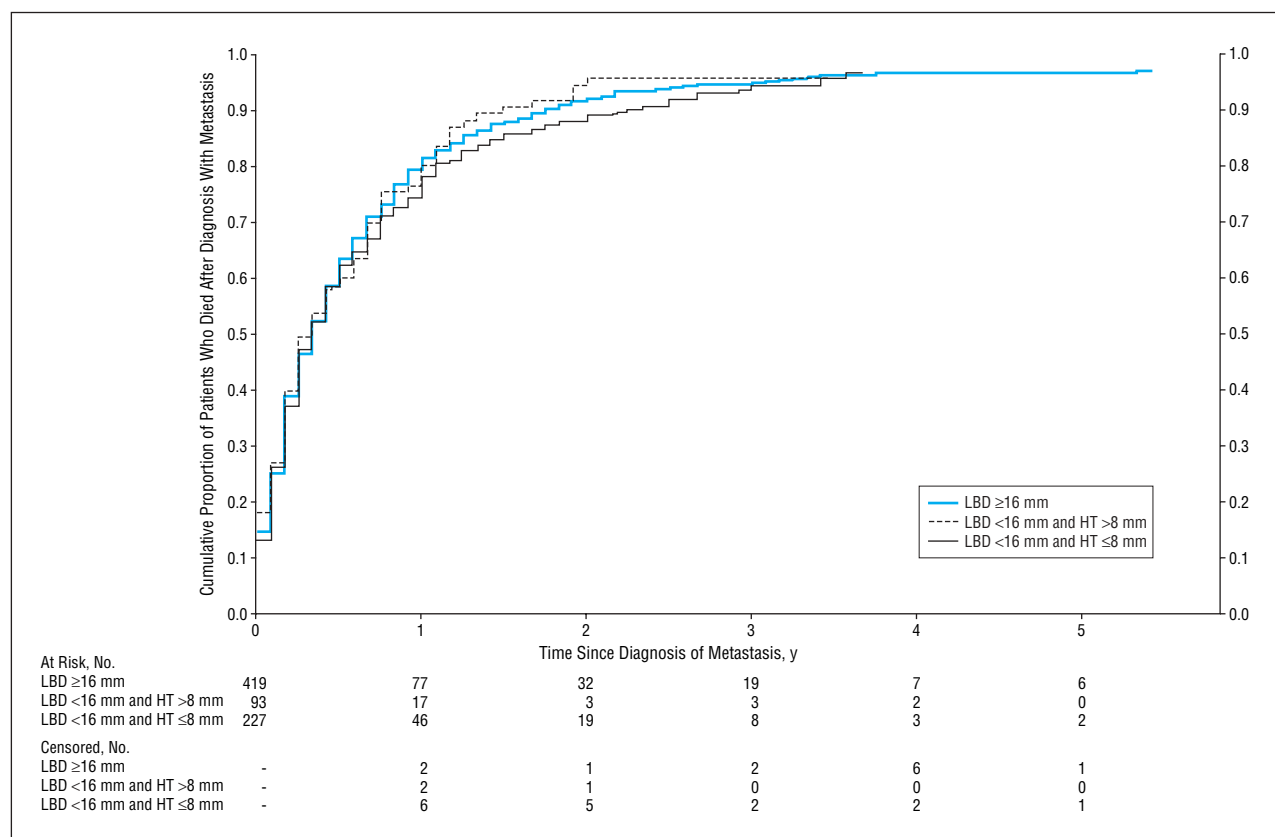


Figure 2. Time to death after diagnosis of first metastasis by baseline tumor size subgroup. HT indicates apical height of tumor; LBD, longest basal diameter of tumor.

Table. Sites of Metastasis by Baseline Tumor Size Subgroup for Patients With Metastasis Reported During Follow-up or at the Time of Death

Site of Metastasis	Tumor Size Subgroup, No. (%)			Total
	LBD ≥16 mm	LBD <16 mm and HT >8 mm	LBD <16 mm and HT ≤8 mm	
Liver	379 (90)	80 (86)	201 (89)	660 (89)
Lung	117 (28)	32 (34)	67 (30)	216 (29)
Bone	72 (17)	14 (15)	37 (16)	123 (17)
Skin or subcutaneous tissue	52 (12)	11 (12)	27 (12)	90 (12)
Lymph node	49 (12)	10 (11)	21 (9)	80 (11)
Brain	24 (6)	2 (2)	11 (5)	37 (5)
Orbit of eye treated for choroidal melanoma	3 (1)	1 (1)	1 (<1)	5 (1)
Spinal cord	3 (1)	1 (1)	4 (2)	8 (1)
Fellow eye or orbit	2 (<1)	0	1 (<1)	3 (<1)
Other	84 (20)	22 (24)	41 (18)	147 (20)
Total	419 (100)	93 (100)	227 (100)	739 (100)

Abbreviations: HT, tumor apical height; LBD, longest basal diameter of tumor.

diagnosis of metastasis; of these, only 4 patients were still alive at the end of COMS clinical follow-up. Their follow-up times since diagnosis of metastasis were 69, 88, 90, and 107 months. Subsequent to the end of COMS patient follow-up, several COMS investigators have reported additional COMS patients surviving with metastasis beyond 5 years.⁸ One patient with metastatic melanoma has been followed up for 10 years since enucleation and 8 years since diagnosis of widespread metas-

tases without intervention for metastasis (L.M. Jampol, oral communication, September 2004).

COMMENT

Given the observed correlation between increased metastasis rate and increased size of choroidal melanoma at time of enrollment and primary treatment, our findings empha-

size the need for earlier diagnosis and treatment of primary choroidal melanoma. In addition, identification of predictors of metastatic disease and development of novel treatment strategies are warranted. We previously described the predictive value of liver function tests.¹⁵ Recent histopathologic analysis has suggested that the presence of epithelial cells and high microvascular density are related to progression of uveal melanoma to metastasis.¹⁶ In addition, survival of patients after diagnosis with metastasis appears inversely correlated with microvascular density.¹⁶ Only a small number of long-term survivors were identified among all COMS patients diagnosed with metastasis. Although all patients had the potential for 5 or more years of follow-up since enrollment in the clinical trials, the end of study data collection may have precluded observation of additional patients with increased survival with metastasis.

We did not observe differences in overall survival after diagnosis of metastasis between patients receiving vs not receiving treatment for metastasis, after controlling for age at time of metastasis. However, interventions for metastatic disease in our trials were selective and optional. Although there have been numerous clinical trials for treatments for patients with cutaneous metastatic melanoma,¹⁷ they typically have excluded patients with choroidal melanoma metastasis. In addition, such patients may not receive metastatic treatment because many medical oncologists will not treat patients who are of advanced age or functioning poorly. Eskelin et al¹⁸ developed a prognostic model and working formulation that categorized predicted median survival as less than 6 months, 6 to 11 months, or greater than or equal to 12 months based on Karnofsky performance status, serum alkaline phosphatase level, and the largest dimension of the largest metastasis. Such a model may be helpful in counseling patients with respect to treatment decisions. Neither Karnofsky performance status nor dimensions of metastases were collected in the COMS and thus could not be assessed in our series. However, measuring alkaline phosphatase level was the liver function test with the best diagnostic attributes among those used in our patient population.¹⁵

Better therapeutic approaches are needed in this group of patients with poor prognosis after diagnosis of melanoma metastasis. Chemotherapy, surgery, and combined modalities have been used with limited success. Transfer of the murine interferon-beta gene as a potential therapy has been explored in mice.¹⁹ Although promising, gene therapy requires further study before widespread use in humans. This subject has been reviewed in a recent comprehensive perspective on choroidal melanoma.²⁰ Given the poor prognosis observed in patients with choroidal melanoma in whom metastasis develops, additional phase 1/2 trials to test new agents and modalities are needed. Our observation is that patients with metastatic melanoma are eager to find and participate in such research.

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